

Midwest Population Genetics 2026

August 21–22, 2026

Friday, August 21

12:00–12:30 pm Registration & Poster Setup (Southwest Lobby)

12:30–12:45 pm Opening Remarks & Welcome

12:45–2:15 pm Session 1: Genomics of Adaptation (Auditorium 1103)

1. Structural variation underlies convergent elevational adaptation — *Sam Mantel, Purdue University (Postdoc)*
2. Adaptive Dynamics of Repeated Local Adaptation in *Amaranthus* — *Libby Polston, University of Chicago (Graduate Student)*
3. Genetic Basis of Melanism in *Drosophila Melanogaster* — *Alexander Kwakye, University of Wisconsin–Madison (Postdoc)*
4. Evolutionary History of Na⁺/H⁺ Antiporter Gene Family — *Patricia Zito, University of Wisconsin–Madison (Graduate Student)*
5. Multiple Colonization Events in *Metrosideros* Polymorpha — *Askhan Shametov, University of Kansas (Graduate Student)*
6. Protein Evolution in Anthocyanin and Carotenoid Loci — *Zoe Hert, Indiana University (Graduate Student)*

2:15–2:45 pm Faculty Spotlight — *Rachel Moran, University of Chicago*

Sex chromosome turnover and mitonuclear conflict drive reproductive isolation

2:45–4:15 pm Poster Session · Last names A–L (Southwest Lobby, coffee & snacks available)

4:15–5:15 pm Session 2: Human Genetics & Complex Traits (Auditorium 1103)

1. Adaptive Introgression in Peruvians Reveals Temporal Waves of Selection — *Susanna Gutierrez, University of Michigan (Postdoc)*
2. Evaluating Methods to Detect Selection Using Ancient DNA — *Sumit Sen, University of Wisconsin–Madison (Graduate Student)*
3. Locus-specific gene-context interactions robustly improve polygenic prediction — *Renee Fonseca, University of Chicago (Graduate Student)*
4. Phenotype Scale in Biobank-based GxE Tests — *Manuela Costantino, University of Chicago (Graduate Student)*

5:15–6:00 pm Travel to Promontory Point

6:00 pm Friday Social & BBQ Dinner (Promontory Point, Hyde Park)

Saturday, August 22

8:30–9:00 am Breakfast & Poster Setup (Southwest Lobby)

9:00–10:00 am Keynote Address — *Angela Hancock, Purdue University*

Finding the molecular basis of adaptation in *Arabidopsis*

10:00–10:20 am Coffee Break (Southwest Lobby)

10:20–11:50 am Session 3: Theory & Inference (Auditorium 1103)

1. Scaling Laws for Site-Frequency Spectrum Under Recent Growth — *Maximillian Newman, University of Chicago (Postdoc)*
2. OOPS: Estimating Mutation Rate From Parent-Offspring Pair — *Trang Nguyen, Indiana University (Graduate Student)*
3. Heavy-Tailed Dispersal and Local Adaptation Revisited — *Sherif Negm, University of Chicago (Graduate Student)*
4. Detecting Selection in the Presence of Gene Flow — *Hank Flury, University of Chicago (Postdoc)*
5. Computing Expected SFS Under Selection and Non-Equilibrium Demography — *Jonathan Wiese, University of Chicago (Graduate Student)*
6. DAFT: Discordant Attachment Frequency Tool — *Sarthak Mishra, Indiana University (Graduate Student)*

11:50 am–12:20 pm Faculty Spotlight — *Arslan Zaidi, University of Minnesota*

A population genetic model of mtDNA transcription

12:20–1:20 pm Lunch (Southwest Lobby)

1:20–2:50 pm Poster Session · Last names M–Z (Southwest Lobby, coffee & snacks available)

2:50–4:35 pm Session 4: Evolutionary Genomics of Non-Model Systems (Auditorium 1103)

1. A waterhemp pangenome resolves repeated herbicide resistance evolution — *Jake Montgomery, University of Chicago (Postdoc)*
2. Genomics of Beetle Evolution via Fusions and Fissions — *Dwayne Tally, Indiana University (Graduate Student)*
3. Mito-nuclear coevolution during stickleback rapid radiation — *Changyue Yu, University of Wisconsin–Madison (Graduate Student)*
4. Kenaitze Nation's Salmon Populations — *Jack Calvery, University of Illinois – Urbana Champaign (Graduate Student)*
5. Mutation Rate Variation in Sexual and Asexual Snails — *Srivarshini Saravanakumar, University of Iowa (Graduate Student)*
6. Deleterious Alleles and Selection in Potato — *Sapphire Coronejo, University of Minnesota (Graduate Student)*
7. Translocations and Deleterious Variation in Imperiled Florida Scrub Jay — *Andrew Mularo, Michigan State University (Postdoc)*

4:35–4:50 pm Judge Deliberation & Raffle

4:50–5:00 pm Awards & Closing